Do Retirees Earn Underlying Market Index Returns?

Another optimistic assumption of classic safe withdrawal rate studies is that retirees are able to precisely earn the underlying index returns. Three truths dispute that idea:

1. Investing costs may reduce returns below the benchmark levels,
2. Many investors may make behavioral and timing mistakes of buying high and selling low, and
3. Actively managed funds do not precisely match the underlying benchmarks.

In reality, many investors may experience investment returns that lag behind the annually rebalanced and indexed portfolios enjoyed by the hypothetical retiree used in the safe withdrawal rate studies.

It is important to consider the impact of account underperformance relative to benchmarks. In Exhibit 5.12, I consider the impact of 1 percent annual account underperformance relative to the index returns, with every other assumption matching my discussion of William Bengen's work on the SAFEMAX. These assumptions include withdrawals taken at the start of the year, annual rebalancing from a 50/50 portfolio of stocks and bonds, a thirty-year retirement, and inflation-adjusted withdrawal amounts. The exhibit shows the pure Bengen results and the results when annual returns are one percentage point less than the underlying indexes.

In both cases, 1966 retirees are the source of our SAFEMAX. With the index returns, a 1966 retiree could sustain withdrawals over thirty years using a 4.03 percent withdrawal rate. With 1 percent underperformance, the SAFEMAX fell by 0.47 percentage points to 3.56 percent. From the perspective of the SAFEMAX, 1 percent underperformance would have resulted in reduced potential spending power of 11.9 percent. Across the historical period, underperformance caused the maximum sustainable withdrawal rates to fall on average by 0.65 percentage points, or 10.9 percent of spending power.